

Database Systems Laboratory – BSDSF23

Lab 02 – Constraints & WHERE Clause in Oracle

Total Marks: 100

Task 1 – Create Tables (20 Marks)

Create COURSES and STUDENTS tables with PRIMARY KEY, NOT NULL, CHECK constraint (credits between 1-5, age ≥ 18), and FOREIGN KEY with ON DELETE CASCADE.

Task 2 – Insert Data (15 Marks)

Insert at least 5 courses and 6 students.

Include two students with same course_id and one with NULL course_id.

Task 3 – Primary Key Rules (10 Marks)

Attempt duplicate student_id and NULL student_id.

Observe errors.

Task 4 – Foreign Key Rules (10 Marks)

Insert invalid course_id (should fail).

Insert valid course_id (should pass).

Task 5 – Cascade Delete (10 Marks)

Delete one course and verify related students are deleted.

Task 6 – UPDATE Operation (5 Marks)

Update credits of one course.

Task 7 – WHERE Clause (20 Marks)

Write queries to:

- Students older than 20
- Students from 'Computer Science'
- Students enrolled in course_id=202
- Students whose name starts with 'A'
- Students with NULL course_id
- Students between age 19 and 23
- Courses with credits=3

Task 8 – Aggregate Functions (10 Marks)

- Count total students
- Count students per course
- Average age
- Maximum credits